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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Operating System Services for Users and Programs

COURSE NAME: Operating System
COURSE CODE:CSA0406

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Introduction

- ▶ Operating System (OS) is system software that acts as an interface between the user, application programs, and computer hardware.
- ▶ It manages system resources such as the CPU, memory, storage, and input/output devices.
- ▶ The OS provides essential services that help users interact with the computer and enable programs to run efficiently.
- ▶ It supports key functions like program execution, file management, memory management, device management, and security.
- ▶ These services improve system performance, reliability, multitasking, and overall user experience.

Services for Users

- ▶ The operating system provides services that make computers easy to use.

Main User Services

- ▶ User Interface (UI) – GUI (Windows) or CLI (Terminal)
- ▶ Program Execution – Runs and terminates applications.
- ▶ File Management – Creates, deletes, reads, and writes files.
- ▶ Communication – Allows users and applications to exchange information.
- ▶ Error Detection – Detects and reports hardware or software errors.
- ▶ **Example:**
When you open Microsoft Word, save a document, and print it, the OS manages all these tasks.

Services for Programs

- ▶ Programs depend on the OS to access system resources.

Main Program Services

- ▶ Memory Management – Allocates memory to running programs.
- ▶ Process Management – Creates, schedules, and terminates processes.
- ▶ I/O Operations – Controls communication with input/output devices.
- ▶ Device Management – Manages printers, keyboards, disks, etc.
- ▶ Protection and Security – Prevents unauthorized access to files and resources.
- ▶ **Example:**
A web browser requests memory, accesses files, and communicates with the internet through OS services.

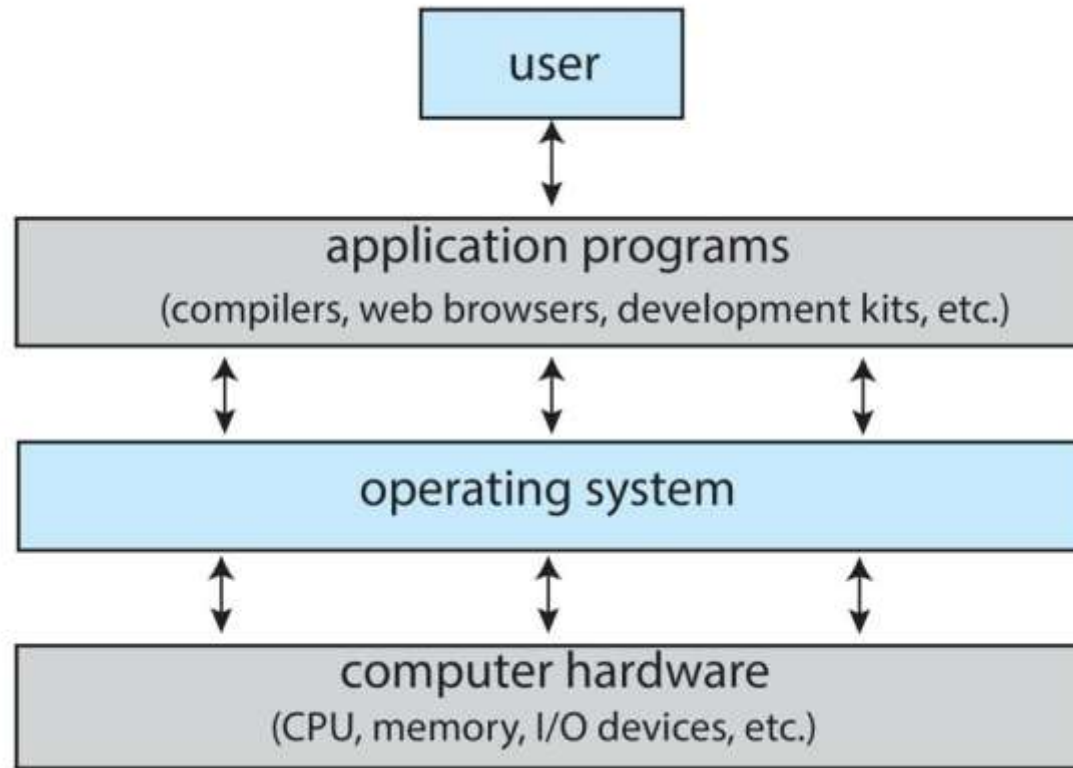
Importance of OS Services

- ▶ Efficient resource management
- ▶ Faster program execution
- ▶ Supports multitasking
- ▶ Protects system and data
- ▶ Provides smooth communication between hardware and software
- ▶ Makes computers user-friendly
- ▶ **Example:** Running MS Word + Chrome + Spotify simultaneously without slowing down the system.

Advantages

- ▶ Faster program execution.
- ▶ Better CPU and memory utilization.
- ▶ Improved system performance.
- ▶ Reduced processing time.
- ▶ Enhanced security and reliability.
- ▶ Easy file and device management.

Basic Architecture of an Operating System



conclusion

- ▶ Operating System is the foundation of every modern computer system.
- ▶ It bridges the gap between users, applications, and hardware.
- ▶ OS services ensure efficient process, memory, file, and device management.
- ▶ These services enhance performance, security, multitasking, and reliability.
- ▶ In conclusion, Operating System services play a vital role in making computer systems efficient, secure, and user-friendly.